

Power Budget

Team Number: 201

Project Name: Garden Buddy

Team Member Name: Rylee Wirt

Version: 02

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150	+1.8 - 5.1V	1	100	100	mA
	2N3904 (NPN transistor)	2N3904	5V	1	200	200	mA
	2N3906 (NPN transistor)	2N3906	5V	1	200	200	mA
	8 Ω Mylar Mini Speaker	CE-2808B36	0 - +5V	1	625	0.625	mA
	LM7805 Linear Regulator	LM7805CT	+7 - 25V	1	5	5	mA
	Regulator quiescent	N/A	+5.0V	1	5	5	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano Board	DM164150	+1.8 - 5.1V	1	100	100	mA
	2N3904 (NPN transistor)	2N3904	40V	1	200	200	mA
	2N3906 (NPN transistor)	2N3906	40V	1	200	200	mA
	8 Ω Mylar Mini Speaker	CE-2808B36	0 - +5V	1	625	0.625	mA
	LM7805 Linear Regulator	LM7805CT	+7 - 25V	1	5	5	mA
	Red LED Indicator	WP710A10SRD/J4	+1.8 - 2.2V	1	2	2	mA
	Regulator/Wiring Allowance	N/A	5.0V rail	1	5	5	mA
Regulator			Subtotal			512.625	mA
			Safety Margin			25%	

			Total Current Required on +5V Rail			640.78125	mA
External Power Source 1							
Power Source 1 Selection	LM7805 Linear Voltage Regulator	LM7805	+7 - 25V	1	1500	1500	mA
	Total Remaining Current Available on +5V Rail					859.21875	mA
Power Rails Connected to External Power Source 1							
	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
	5V Wall Adapter (Regulated, Certified)	Generic UL/CE	100 - 240VAC	+5.0V	2000	10000	mA
on External Power Source 1							
	N/A	N/A	N/A	N/A	N/A	N/A	mA
Calculated Battery Life	+5V Regulator	LM7805	+7 - 25V	1	1500	1500	mA
	N/A	N/A	N/A	N/A	N/A	N/A	mA
						8500	mA
Notes:	Component Name	Part Number	Supply		Capacity	Required	
External Supply Voltage	N/A	N/A	N/A	N/A	N/A	N/A	
in your design (e.g., a base unit and remote unit) then						Battery Life	#VALUE!
							hours